

- 3 (a) Two light identical springs are joined in series and a 120 g bar magnet is hung from the lower end as shown in Fig. 3.1. The effective spring constant of the combined springs is 11 N m^{-1} .

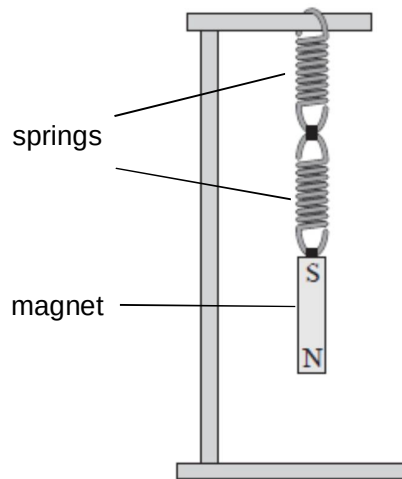


Fig. 3.1

The bar magnet is displaced a small distance vertically downward from its equilibrium position and released. It then oscillates in simple harmonic motion.

- (i) By considering the forces on the magnet at its equilibrium position, and also at a small displacement x downward from its equilibrium position, show that the oscillation is a simple harmonic motion where the acceleration a is given by the equation

$$a = -92 x$$

Let downwards be positive.

Let e be the extension of the spring when the magnet is at its equilibrium position.

At the equilibrium position,

$$ke = mg \quad \text{----- (1)}$$

M1

At a displacement x downward from its equilibrium position,

$$mg - k(x + e) = ma$$

$$mg - kx - ke = ma \quad \text{----- (2)}$$

M1

Sub (1) into (2):

$$-kx = ma$$

$$a = -(k/m)x$$

$$a = -[11/(120 \times 10^{-3})]x$$

$$a = -91.667x$$

$$a = -92x \quad \text{(shown)}$$

M1

M1

A0

(ii) Hence calculate the frequency at which the magnet oscillates.

$$\text{frequency} = \text{-----} \text{ Hz} \quad [2]$$

$$a = -92x$$

Comparing with the SHM equation, $a = -\omega^2 x$,

$$\omega^2 = 92$$

$$\omega = \sqrt{92}$$

$$2\pi f = \sqrt{92}$$

$$f = 1.5266 = 1.5 \text{ Hz}$$

M1

A1

(b) Two systems in part **(a)** are now suspended with the North pole of their magnets located inside a coil of wire. The two coils are connected together with conducting leads as shown in Fig. 3.2.

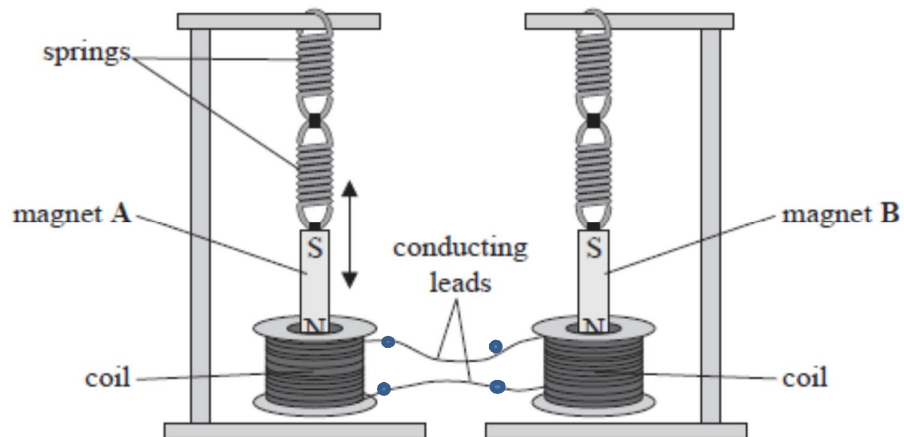


Fig. 3.2

Magnet A is displaced so that it oscillates vertically. The North pole of magnet A moves into and out of the coil of wire with simple harmonic motion. As this motion continues, magnet B starts to oscillate. The amplitude of oscillation of magnet B increases over time.

Explain why magnet B starts to oscillate with an increasing amplitude.

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[5]

As magnet A moves in and out of the coil, its coil experiences a change of magnetic flux density and hence a change in magnetic flux linkage. B1

By Faraday's law, this change in magnetic flux linkage induces an electromotive force in the coil which then causes a current in both coils (since the coils are connected to each other). B1

The alternating current in the second coil creates a periodic force on magnet B and driving it into a forced oscillation. B1

Since both spring-magnet systems have the same natural frequency of oscillation, B1

resonance occurs and thus there is a maximum transfer of energy from magnet A to magnet B and magnet B oscillates with increasing amplitude. B1